Applied Regression Interpretation

Gov 51: Data Analysis and Politics

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The Big Picture

Women earn 84 cents for every dollar men earn

Where does that number come from?

And what does it actually mean?

Today: one dataset, four skills, building complexity

Interpret → **Compare** → **Interact** → **Predict**

526 workers from the 1976 CPS — every number today is a fact about these people

The Current Population Survey is the gold standard for labor data

- **What:** Monthly survey of ∼60,000 U.S. households
- **Who runs it:** Bureau of Labor Statistics + Census Bureau
- **How:** Interviewers visit or call sampled households
- **Why it exists:** Official source for unemployment rate, earnings, labor force participation
- Every jobs report you see in the news comes from the CPS

Wooldridge extracted 526 workers from the 1976 CPS

- 1976: pre-Equal Pay enforcement, large raw gender gaps
- Clean, real data — every number we compute today is a fact about these 526 people

$$n = 526 \cdot 274 \text{ men, } 252 \text{ women}$$

Six variables, one dataset, the whole lecture

Variable	Type	Description
`wage`	Continuous	Hourly wage ($/hr)
`educ`	Continuous	Years of education
`exper`	Continuous	Years of work experience
`tenure`	Continuous	Years at current job
`female`	Binary (0/1)	= 1 if woman
`married`	Binary (0/1)	= 1 if married

Model 1: The Raw Gap

Model 1 starts with the population regression

$$\text{wage}_i \quad = \quad \beta_0 \; + \; \beta_1 \; \cdot \; \text{female}_i \; + \; \epsilon_i$$

- wage_i — the **outcome** (*dependent variable*): hourly wage for person i
- female_i — the **predictor** (*independent variable*): equals 1 if woman, 0 if man
- β_0, β_1 — **unknown parameters** we want to learn
- ϵ_i — the **error term**: everything else that affects wages

OLS minimizes the sum of squared residuals

$$\min_{\hat{\beta}_0, \hat{\beta}_1} \sum_{i=1}^{n} \hat{e}_i^2 \quad = \quad \sum_{i=1}^{n} \left(Y_i \; - \; \hat{\beta}_0 \; - \; \hat{\beta}_1 X_i\right)^2$$

Choose $\hat{\beta}_0$ and $\hat{\beta}_1$ to make the residuals as small as possible

The solution gives us fitted values for every worker

$$\widehat{\text{wage}}_i \quad = \quad \hat{\beta}_0 \; + \; \hat{\beta}_1 \; \cdot \; \text{female}_i$$

Plug in female $= 0$ or 1 and you get a predicted wage

Women earn \$2.51 less per hour than men on average

	Model 1
Intercept ($\hat{\beta}_0$)	7.10
	(0.21)
Female ($\hat{\beta}_1$)	**−2.51**
	(0.30)
R^2	0.116
n	526

Five numbers. By the end of today, you will know what every one of them means.

What do 7.10 and −2.51 mean?

$$\widehat{\text{wage}}_i = \underbrace{7.10}_{\hat{\beta}_0} + \underbrace{(-2.51)}_{\hat{\beta}_1} \cdot \text{female}_i$$

If female = 0 (men), what is $\hat{Y}$?

If female = 1 (women), what is $\hat{Y}$?

So what is $\hat{\beta}_1$ measuring?

$\hat{\beta}_0$ is the male mean; $\hat{\beta}_1$ is the gap

- Men (female = 0): $\hat{Y} = 7.10 \quad \Rightarrow \quad \hat{\beta}_0 = \overline{\text{wage}}_{\text{men}}$
- Women (female = 1): $\hat{Y} = 7.10 - 2.51 = 4.59 \quad \Rightarrow \quad \hat{\beta}_0 + \hat{\beta}_1 = \overline{\text{wage}}_{\text{women}}$
- Subtract: $\hat{\beta}_1 = 4.59 - 7.10 = -2.51$

With binary X: intercept = baseline mean, slope = difference in means

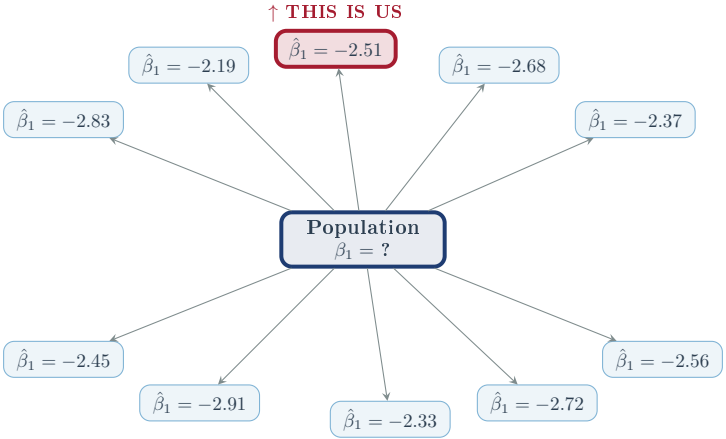
What is the (0.30) underneath the coefficient?

	Model 1
Female ($\hat{\beta}_1$)	-2.51
	(0.30) ← **what is this?**

We got -2.51 from 526 workers.

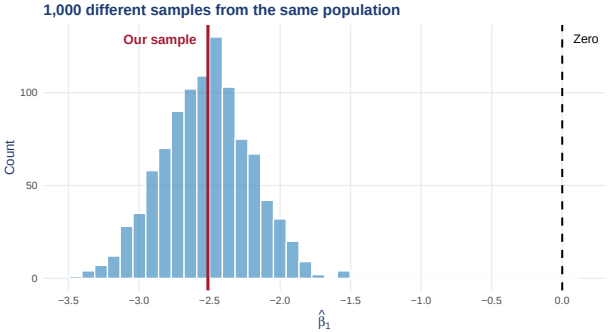
What if we had surveyed a *different* 526 workers?

Our sample is one of many we could have drawn



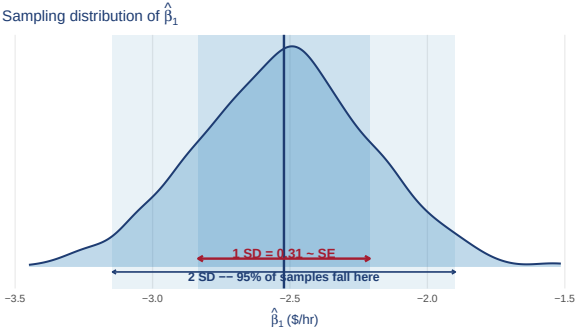
Every sample gives a different $\hat{\beta}_1$ — all are estimates of the same β_1

Here are 1,000 different samples from the same population



The **standard error** measures the width of this distribution: SE = 0.30

The SE is the standard deviation of this distribution



$$\text{SE}(\hat{\beta}_1) \quad \approx \quad \text{SD of all possible } \hat{\beta}_1\text{'s} \quad \approx \quad 0.30$$

β_1 is the truth; $\hat{\beta}_1$ is our guess from data

The population (all U.S. workers)

$\text{wage}_i = \beta_0 + \beta_1 \cdot \text{female}_i + \epsilon_i$

- β_1 = true average wage gap
- This is what we **want to know**
- Not a causal claim — just a difference

Our sample (526 workers)

$\hat{\beta}_1 = -2.51$

- One draw from that distribution
- From *this particular* sample
- The SE tells us how much it bounces

Hats mean "estimated from this sample" — not the true population value

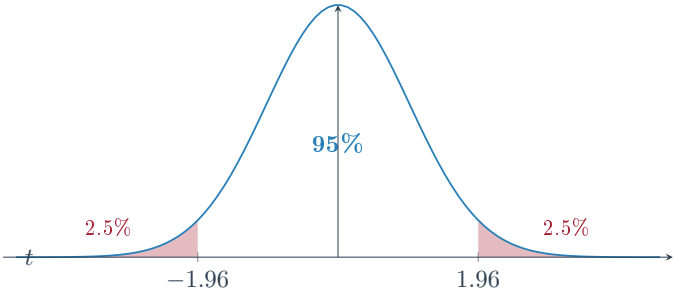
The t-statistic asks: could the true gap be zero?

- **Thought experiment:** Suppose $\beta_1 = 0$ (no real wage gap in the population)
- We got $\hat{\beta}_1 = -2.51$ — how surprising is that?
- Measure surprise in units of SE:

$$t = \frac{\hat{\beta}_1}{\text{SE}(\hat{\beta}_1)} = \frac{-2.51}{0.30} = -8.28$$

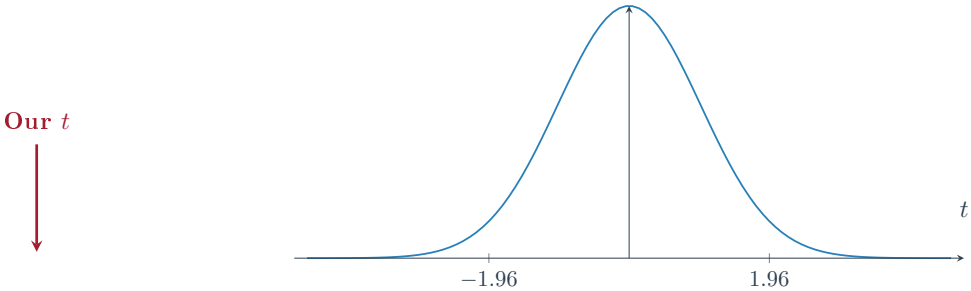
Our estimate is **8.28 standard errors** away from zero

The t-distribution shows which values are plausible under H_0



- Blue: 95% of t-statistics fall between ± 1.96 if $\beta_1 = 0$
- Red tails: only 5% land here — these are "unlikely under H_0"

Our $t = -8.28$ is not even on the chart



The p-value is the chance of landing this far out if $\beta_1 = 0$ — here it is ≈ 0

$p < 0.001$ means: if the true gap were zero, you'd essentially never see $t = -8.28$

From hypothesis test to confidence interval

- We rejected $\beta_1 = 0$ — zero is implausible
- But which values of β_1 **are** plausible?
- **Answer:** values we *cannot* reject — within 1.96 SEs of $\hat{\beta}_1$

$$95\% \text{ CI} \;=\; \hat{\beta}_1 \pm 1.96 \times \text{SE} \;=\; -2.51 \pm 1.96(0.30)$$

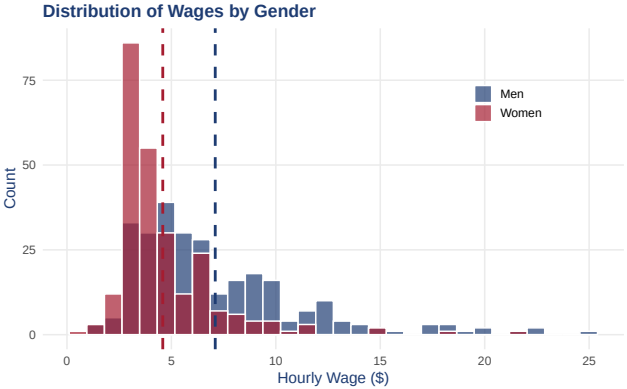
$$95\% \text{ CI} \quad = \quad (-3.10,\ -1.92)$$

We are 95% confident the true gap is between $-\$3.10$ and $-\$1.92$

- **Correct:** If we repeated this sampling 100 times, ∼95 of those intervals would contain β_1
- **Wrong:** "There is a 95% chance β_1 is in this interval"
  - β_1 is fixed — it's either in there or it isn't
- Zero is **not** in $(-3.10,\ -1.92)$ — consistent with rejecting H_0

You will need **1.96** for confidence intervals on the exam

Is knowing someone's gender enough to predict their wage?



Look at the overlap. How much wage variation does gender explain?

$R^2 = 0.116$ means gender alone explains 12% of wage variation

- 88% of variation is *within* gender groups
- Some men earn very little; some women earn a lot
- Gender matters, but it is far from the whole story

$$R^2 = 1 - \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\sum \hat{e}_i^2}{\sum (Y_i - \bar{Y})^2} = 0.116$$

Turn to your neighbor

"The wage gap is \$2.51."

What does that number **not** tell you?

Model 2: Adding Covariates

Model 2 adds education, experience, and tenure

$$\widehat{\text{wage}}_i = \hat{\beta}_0 + \hat{\beta}_1 \text{female}_i + \hat{\beta}_2 \text{educ}_i + \hat{\beta}_3 \text{exper}_i + \hat{\beta}_4 \text{tenure}_i$$

- Same outcome, same treatment variable
- Now we **hold constant** education, experience, and tenure
- This asks a *different question* than Model 1

Before we look: will the female coefficient get bigger or smaller?

Model 1: $\hat{\beta}_{\text{female}} = -2.51$

We are about to add education, experience, and tenure.

	Men	**Women**
Mean education	12.8 yrs	12.3 yrs
Mean experience	17.6 yrs	16.4 yrs
Mean tenure	6.5 yrs	3.6 yrs

Will the magnitude of $\hat{\beta}_{\text{female}}$ increase or decrease? Why?

The gender gap shrinks from \$2.51 to \$1.81 after adding controls

	Model 1	Model 2
Intercept	7.10	−1.57
	(0.21)	(0.72)
Female	−2.51	**−1.81**
	(0.30)	(0.26)
Education		0.57
		(0.05)
Experience		0.03
		(0.01)
Tenure		0.14
		(0.02)
R^2	0.116	**0.364**
n	526	526

Every coefficient has a "holding constant" interpretation

- **Female:** $-\$1.81$/hr, holding education, experience, and tenure constant
- **Education:** $+\$0.57$/hr per additional year, holding gender, experience, and tenure constant
- **Experience:** $+\$0.03$/hr per additional year, holding gender, education, and tenure constant
- **Tenure:** $+\$0.14$/hr per additional year, holding gender, education, and experience constant

"Holding constant" changes the question we are asking

Model 1

$\hat{\beta}_{\text{female}} = -2.51$

Gap between the *average* man and the *average* woman

Model 2

$\hat{\beta}_{\text{female}} = -1.81$

Gap between a man and woman with the *same* education, experience, and tenure

Adding covariates changes what $\hat{\beta}_1$ means

R^2 tripled — the model explains 3× more variation

- Model 1: $R^2 = 0.116$
- Model 2: $R^2 = 0.364$
- Education, experience, and tenure explain a lot of wage variation
- The SE on Female *shrank*: $0.30 \to 0.26$
  - More precise estimate when the model fits better

Turn to your neighbor: plug-in prediction

Using Model 2:

$$\hat{Y} = -1.57 + (-1.81)\text{female} + 0.57 \cdot \text{educ} + 0.03 \cdot \text{exper} + 0.14 \cdot \text{tenure}$$

Predict hourly wage for:

Profile: educ = 16, exper = 10, tenure = 5

1. A woman with this profile: $\hat{Y} = ?$

2. A man with this profile: $\hat{Y} = ?$

3. What is the gap?

Plug-in prediction: the gap is always $\hat{\beta}_{\textbf{female}}$

- **Man** (female $= 0$):
  $\hat{Y} = -1.57 + 0(-1.81) + 0.57(16) + 0.03(10) + 0.14(5) = \$8.54$
- **Woman** (female $= 1$):
  $\hat{Y} = -1.57 + 1(-1.81) + 0.57(16) + 0.03(10) + 0.14(5) = \$6.72$
- **Gap**: $6.72 - 8.54 = -\$1.81 = \hat{\beta}_{\text{female}}$

With no interaction, the gap is the same for every profile

Model 3: Interactions

Does marriage affect everyone's wages the same way?

Think about this:

- In 1976, who was more likely to be the primary earner in a married household?
- If employers perceive married men as "more stable," what happens to their wages?
- If employers perceive married women as "likely to leave," what happens?

*How would you **model** this — one slope for marriage, or two?*

Model 3 asks: does marriage affect wages differently for men and women?

$$\widehat{\text{wage}}_i = \hat{\beta}_0 + \hat{\beta}_1 \text{female}_i + \hat{\beta}_2 \text{married}_i + \hat{\beta}_3 (\text{female} \times \text{married})_i + \cdots$$

- $\hat{\beta}_2$ = effect of marriage **for men** (when female = 0)
- $\hat{\beta}_2 + \hat{\beta}_3$ = effect of marriage **for women**
- $\hat{\beta}_3$ = *difference* in marriage effect across genders

Marriage boosts men's wages by \$1.82 but *reduces* women's

	Model 1	Model 2	Model 3
Intercept	7.10	−1.57	−2.39
	(0.21)	(0.72)	(0.73)
Female	−2.51	−1.81	−0.31
	(0.30)	(0.26)	(0.41)
Married			1.82
			(0.40)
Female × Married			**−2.40**
			(0.53)
Education		0.57	0.55
		(0.05)	(0.05)
Experience		0.03	0.02
		(0.01)	(0.01)
Tenure		0.14	0.13
		(0.02)	(0.02)
R^2	0.116	0.364	**0.392**
n	526	526	526

The marriage premium: \$1.82 for men, −\$0.57 for women

Men (female = 0)

Marriage effect = $\hat{\beta}_2$
= +\$1.82/hr

Married men earn significantly more

Women (female = 1)

Marriage effect = $\hat{\beta}_2 + \hat{\beta}_3$
= $1.82 + (-2.40) = -\$0.57$/hr

Married women earn slightly less

$\hat{\beta}_3 = -2.40$ captures the *difference* in marriage premiums

Turn to your neighbor: 2×2 predictions

Using Model 3 with educ = 12, exper = 10, tenure = 5:

$$\hat{Y} = -2.39 + (-0.31)\text{female} + 1.82 \cdot \text{married} + (-2.40)\text{female} \times \text{married}$$
$$+0.55 \cdot \text{educ} + 0.02 \cdot \text{exper} + 0.13 \cdot \text{tenure}$$

Fill in the table:

	Single	**Married**
Men	?	?
Women	?	?

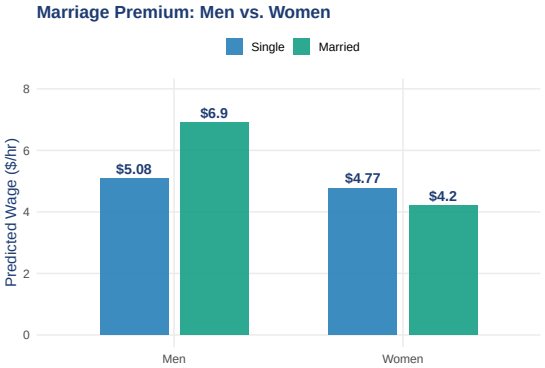
Which group earns the most? The least?

Married men earn the most; married women earn the least

	Single	Married
Men	$5.08	$6.90
Women	$4.77	$4.20

Marriage premium:

Men: +$1.82 Women: −$0.57



Putting It All Together

The gap shrinks as you add controls:
$-\$2.51 \rightarrow -\$1.81 \rightarrow -\$0.31$

Adding controls changes how large the gap is — and what it *means*

Each model answers a different question

Model	What $\hat{\beta}_{\text{female}}$ measures
1	Raw gap: average man vs. average woman
2	Adjusted gap: same education, experience, tenure
3	Gap for single workers with same human capital

The "wage gap" depends on which question you ask

Polynomials

The 30th year of experience matters less than the 5th

$$\widehat{\text{wage}}_i = \hat{\beta}_0 + \hat{\beta}_1 \text{female}_i + \hat{\beta}_2 \text{educ}_i + \hat{\beta}_3 \text{exper}_i + \hat{\beta}_4 \text{exper}_i^2 + \hat{\beta}_5 \text{tenure}_i$$

- $\hat{\beta}_3 = 0.205, \quad \hat{\beta}_4 = -0.004$
- Negative $\hat{\beta}_4$: each additional year adds *less* than the last

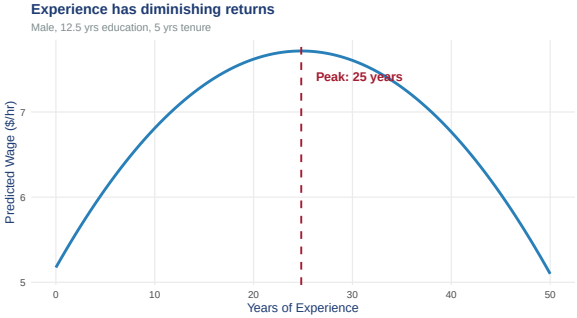
R: `lm(wage ~ female + educ + exper + I(exper^2) + tenure)`

The marginal effect depends on where you start

$$\frac{\partial\,\widehat{\text{wage}}}{\partial\,\text{exper}} = \hat{\beta}_3 + 2\hat{\beta}_4 \cdot \text{exper} = 0.205 - 0.008 \cdot \text{exper}$$

Experience	Marginal effect (\$/hr)
5 years	+\$0.16
15 years	+\$0.08
25 years	≈ \$0.00
35 years	−\$0.08

Wages peak at about 25 years of experience



$$\text{exper}^* = \frac{-\hat{\beta}_3}{2\hat{\beta}_4} = \frac{-0.205}{2(-0.004)} \approx 25 \text{ years}$$

Quadratics are still OLS — exper^2 is just another column

	X_1	X_2	X_3	$\cdots$
Worker 1	5	25	12	
Worker 2	20	400	16	
Worker 3	8	64	10	
	exper	exper^2	educ	

exper^2 is a **mechanical transformation** — OLS doesn't know it's a square

Five Ways to Read a Coefficient

The interpretation depends on what X and Y are

Same $\hat{\beta}_1$ formula. Five different English sentences.

Y **can be:**	continuous (wage) or binary (highwage) or logged
X **can be:**	continuous (educ) or binary (female, college)

The units of $\hat{\beta}_1$ change with the units of X and Y

Continuous → Continuous: \$0.54/hr per year of education

$$\widehat{\text{wage}}_i \quad = \quad -0.90 \; + \; 0.54 \; \cdot \; \text{educ}_i$$

$$\hat{\beta}_1 = 0.54$$

One more year of education → +\$0.54/hr

Units of $\hat{\beta}_1$: dollars per year

Continuous → Log: +8.3% per year of education

$$\widehat{\log(\text{wage})}_i \quad = \quad 0.58 \; + \; 0.083 \; \cdot \; \text{educ}_i$$

$$\hat{\beta}_1 = 0.083$$

One more year of education → +8.3% higher wages

- **Rule:** $\hat{\beta}_1 \times 100 =$ percent change in Y
- Works well when $\hat{\beta}_1$ is small

Binary → Continuous: women earn \$2.51 less per hour

$$\widehat{\text{wage}}_i \quad = \quad 7.10 \; + \; (-2.51) \; \cdot \; \text{female}_i$$

$$\hat{\beta}_1 = -2.51$$

$$\hat{\beta}_1 \quad = \quad \bar{Y}_{\text{women}} - \bar{Y}_{\text{men}} \quad = \text{difference in group means}$$

You already know this one from Model 1

Binary → Log: women earn ∼40% less

$$\widehat{\log(\text{wage})}_i \quad = \quad 1.81 \; + \; (-0.40) \; \cdot \; \text{female}_i$$

$$\hat{\beta}_1 = -0.40$$

$$\text{Approximate: } \hat{\beta}_1 \; \times \; 100 \quad = \quad -40\%$$

$$\text{Exact: } (e^{\hat{\beta}_1} \; - \; 1) \; \times \; 100 \quad = \quad -32.8\%$$

When $|\hat{\beta}_1|$ is large, the approximation overshoots

Binary → Binary: college grads are 42pp more likely to earn above median

- **Define:** `highwage` $= 1$ if wage $>$ median; `college` $= 1$ if educ ≥ 16

$$\widehat{\text{highwage}}_i = 0.35 + 0.42 \cdot \text{college}_i$$

$$\hat{\beta}_1 = 0.42$$

College graduates are **42 percentage points** more likely to earn above median

This is the **Linear Probability Model** (LPM): OLS with a binary Y

Five specifications, five interpretations

Y	**X**	$\hat{\beta}_1$	**Interpretation**
wage	educ (cont.)	0.54	+\$0.54/hr per year
log(wage)	educ (cont.)	0.083	+8.3% per year
wage	female (binary)	−2.51	Women earn \$2.51 less
log(wage)	female (binary)	−0.40	Women earn ∼40% less
highwage	college (binary)	0.42	+42 pp more likely

Same formula. Same estimator. The **units** change with X and Y.

Turn to your neighbor: interpret every coefficient

Full model:

$$\widehat{\log(\text{wage})}_i = 0.42 - 0.30\,\text{female}_i + 0.088\,\text{educ}_i + 0.005\,\text{exper}_i + 0.017\,\text{tenure}_i$$

1. Female: -0.30 means ...
2. Education: 0.088 means ...
3. Experience: 0.005 means ...
4. Tenure: 0.017 means ...

Hint: Y is logged, so $\hat{\beta} \times 100 =$ percent change

Prediction and Overfitting

Today's plan

We are not doing causal inference right now

No treatments, no counterfactuals — just: can we predict Y well?

1. **OLS prediction basics** — covariates, polynomials, R^2
2. **Recognizing overfitting** — when "better fit" makes predictions worse
3. **Train/test splits** — honest evaluation of prediction accuracy (review from earlier)
4. **After spring break:** machine learning methods designed for prediction

People predict things all the time

High stakes

- Will this patient respond to treatment?
- Which counties will swing in the next election?
- Where will the next violent crime occur?
- How many ER visits will flu season bring?
- What will GDP growth be next quarter?

Everyday stakes

- Who should I start in fantasy football?
- Will this movie bomb at the box office?
- How much should I list my house for?
- Which team covers the spread Saturday?
- How long will my commute take tomorrow?

The common thread: predicting what has not yet happened

All of them need to predict **outcomes they have not yet observed**

Not "what happened?" but "what will happen?"

You already know a prediction machine: OLS

$$\hat{Y}_i \;=\; \hat{\beta}_0 \;+\; \hat{\beta}_1 X_{1i} \;+\; \hat{\beta}_2 X_{2i} \;+\; \cdots \;+\; \hat{\beta}_k X_{ki}$$

- The $\hat{\beta}$'s describe patterns *in the sample*
- But OLS draws a **line** — and lines extend beyond the data
- Plug in *any* values of X and you get a $\hat{Y}$, even for combinations nobody in the data has

That is prediction: estimating Y for observations the model has never seen

The intercept is already a prediction

$$\hat{\beta}_0 = \text{predicted } Y \text{ when } \textbf{every } X = 0$$

Model	**Intercept meaning**
Wages $\sim$ Female	Mean wages for men (the $X = 0$ group) — sensible
Wages $\sim$ Female + Exper	Mean wages for men with zero experience — plausible
Wages $\sim$ Female + Exper + Age	Mean wages for a zero-year-old man with zero experience — impossible

The intercept predicts "leftward" to the Y-axis
— even when no one in the data lives there

Description vs. prediction

Description

Goal: patterns *in* the data

X**'s:** observed in the sample

$\hat{Y}$**:** summarizes what we have

Prediction

Goal: Y for *new* observations

X**'s:** may be unseen combinations

$\hat{Y}$**:** a bet on the unseen

Same model, same $\hat{\beta}$'s — the difference is
whether the X's are in-sample or out-of-sample

Ames, Iowa: 2,930 houses, 74 variables, one goal

Can we predict a house's sale price?

Outcome:	Sale price ($12K–$755K)
Predictors:	Square footage, year built, garage size, lot area, bedrooms, bathrooms, pool, ...
Question:	How many variables should we use?

R^2 looks great when you add more variables

	1 var	**4 vars**	**20 vars**	**259 vars**
Train R^2	0.52	0.83	0.86	**0.92**

R^2 **always** goes up (or stays the same) when you add variables

So why not add *everything*?

With enough variables, $R^2 = 1.00$ is guaranteed

Variables (k)	Observations (n)	Train R^2
1	2,051	0.52
4	2,051	0.83
20	2,051	0.86
259	2,051	0.92
$k \geq n$	2,051	**1.00**

If $k \geq n$, OLS can **perfectly** fit every data point.
$R^2 = 1.00$ is not learning — it is memorization.

Think about this

You build a model with 2,051 variables on 2,051 houses.

$R^2 = 1.00$. Every prediction is perfect.

Your friend lists a new house for sale.

How confident are you in your prediction of its price?

R^2 only measures how well you fit THIS dataset

- R^2 rewards **memorization**
- A model with 259 terms can memorize patterns in the noise
- **The real question:** can the model predict *new* houses it hasn't seen?

A perfect fit on your data might be a terrible fit on new data

On new data, R^2 can go *negative*

Imagine: you fit a model on your dataset, then a *new* batch of houses arrives

- **On the original data:** $R^2 \geq 0$ always — OLS guarantees SSR $\leq$ SST
- **On new data:** if predictions are worse than guessing $\bar{Y}$, then SSR $>$ SST

$$R^2_{\text{new}} = 1 - \frac{\text{SSR}_{\text{new}}}{\text{SST}_{\text{new}}} < 0 \quad \text{when SSR} > \text{SST}$$

But is $R^2_{new} < 0$ possible? Both SSR and SST are sums of squares — both positive. What would make the ratio exceed 1?

Yes — here it is, with our Ames data

630-coefficient model (all pairwise interactions) trained on 2,051 houses, then applied to 879 *different* houses

$$\begin{aligned}
\text{SSR}_{\text{new}} &= \textstyle\sum_{i \in \text{new}} (Y_i - \hat{Y}_i)^2 \;\approx\; 32{,}648 \text{ billion} \\
\text{SST}_{\text{new}} &= \textstyle\sum_{i \in \text{new}} (Y_i - \bar{Y})^2 \;\approx\; 4{,}973 \text{ billion} \\
\frac{\text{SSR}}{\text{SST}} &= \frac{32{,}648}{4{,}973} = 6.57 \\
R^2_{\text{new}} &= 1 - 6.57 = \mathbf{-5.57}
\end{aligned}$$

The model's squared errors are **6.5× larger** than just guessing \$181,515 for every house

The fix: train on some data, test on the rest

- **Step 1:** Split data — **training set** (70%) and **test set** (30%)
- **Step 2:** Fit the model on training data only
- **Step 3:** Predict on test data the model has *never seen*
- **Step 4:** Measure prediction error

$$\text{RMSE} = \sqrt{\frac{1}{n_{\text{test}}} \sum_{i \in \text{test}} \left(Y_i - \hat{Y}_i\right)^2}$$

RMSE = Root Mean Squared Error — same units as Y (dollars)

Adding good predictors helps everywhere

	1 variable	4 variables
Train RMSE	\$57,000	\$34,000
Test RMSE	\$56,000	\$36,000

Both train *and* test error improve
— the model learns real patterns

R^2 rising = better fit. RMSE falling = better fit. Two ways to say the same thing.

Throwing in everything: test predictions go bananas

	1 var	**4 vars**	**20 vars**	**259 vars**
Train R^2	0.52	0.83	0.86	0.92
Test R^2	0.45	0.77	**0.81**	0.56
Train RMSE	\$57K	\$34K	\$30K	\$23K
Test RMSE	\$56K	\$36K	\$33K	**\$50K**

Train RMSE keeps falling. Test RMSE **rises** for the 259-variable model.

What does $50K of prediction error mean in practice?

- **The 259-variable model** promised $23K average error (training RMSE)
- **On new houses,** the actual error is $50K — more than double
- A buyer relying on this model could overpay by $50,000
- A seller could underprice by $50,000

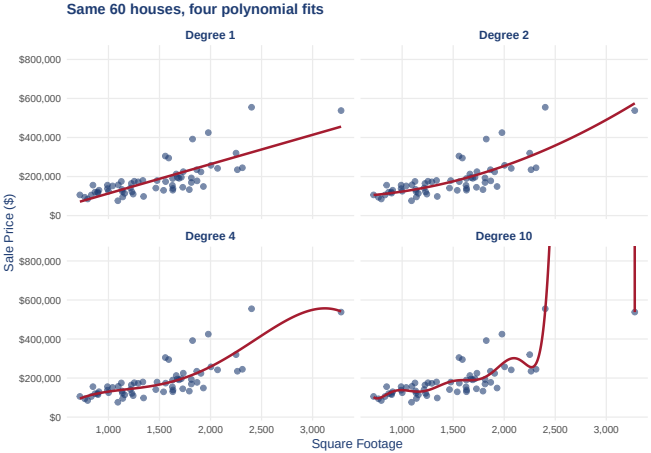
The model looked great on paper. It failed when it mattered.

Why does this happen? The model memorized noise.

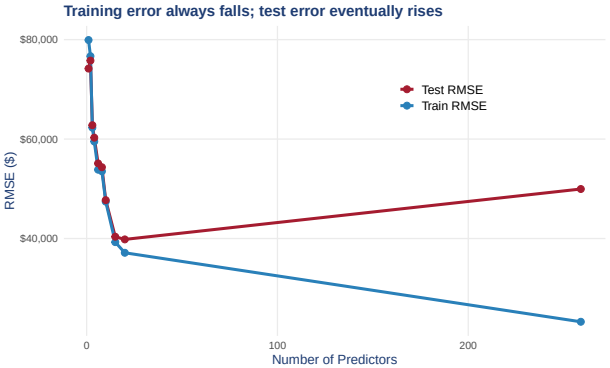
- With 259 variables and 2,051 houses, the model has **one parameter for every 8 observations**
- It learns real patterns (square footage matters) *and* fake patterns (noise in the training data)
- On new data, the real patterns generalize — the fake ones do not

Overfitting = fitting signal *and* noise, instead of just signal

Degree 10 chases individual houses instead of learning the pattern



Training error always falls; test error eventually rises



A model that memorizes training data **fails** on new data

More variables $\neq$ better model

- **Prediction:** kitchen-sink models memorize noise $\rightarrow$ terrible forecasts
- **Causal inference:** throwing in every control variable can amplify bias
- **Lesson:** choose predictors with purpose, not volume

More variables $\neq$ better model — this is true for prediction *and* causal inference

Can we find the *optimal* prediction model?

What we know so far:

- Too few variables → underfitting (misses real patterns)
- Too many variables → overfitting (memorizes noise)
- Test RMSE reveals the sweet spot — but only for models we tried

Open question: is there a principled way to *search* for the best model, instead of guessing?

Why $k \geq n$ breaks OLS: the algebra

OLS requires inverting $(X'X)$:

$$\hat{\boldsymbol{\beta}} \quad = \quad (X'X)^{-1}X'Y$$

- X is $n \times k$ — when $k \geq n$, the columns are linearly dependent
- $(X'X)$ becomes **singular** — no unique inverse exists
- Infinitely many $\hat{\boldsymbol{\beta}}$ solve the normal equations

With $k \geq n$, OLS has **no unique solution**. The model can perfectly fit the data in infinitely many ways.

Coming soon: penalized regression & cross-validation

The machine learning solution:

1. **Penalize complexity** — add a cost for large coefficients (Ridge, LASSO)
   Forces the model to keep only variables that earn their place
2. **Cross-validation** — systematically rotate which data is "train" vs. "test"
   Finds the penalty strength that minimizes out-of-sample error
3. **Result:** a principled, automated search for the best-predicting model

Not today — but this is where we're headed next

Exam Preparation

Five skills to master before the exam

1. **Interpret coefficients in context**
2. **Compute $\hat{Y}$ by plug-in**
3. **Compare models side-by-side**
4. **Read interactions and build 2×2 tables**
5. **Recognize overfitting**

Study the review guide — the exam comes from it

- Not everything on the review will appear
- But the exam is drawn *from* the review

Practice plug-in predictions, interpret every number in a regression table, build 2×2 tables

- **Cheat sheet:** 2 pages, front and back, handwritten or typed
- **Exam 1:** Thursday, March 12 — 75 minutes

Three papers you are responsible for

1. **Card et al.** — anti-immigration speeches and media framing
2. **Broockman, Kalla & Aronow (2015)** — irregularities in the LaCour study
3. **Broockman & Kalla (2016)** — transgender rights canvassing experiment

I reserve the right to ask you **anything** about these three papers

We spent two full lectures on these — use your lecture notes and slides to review

Every number in a regression table has a precise English meaning — your job is to find it